

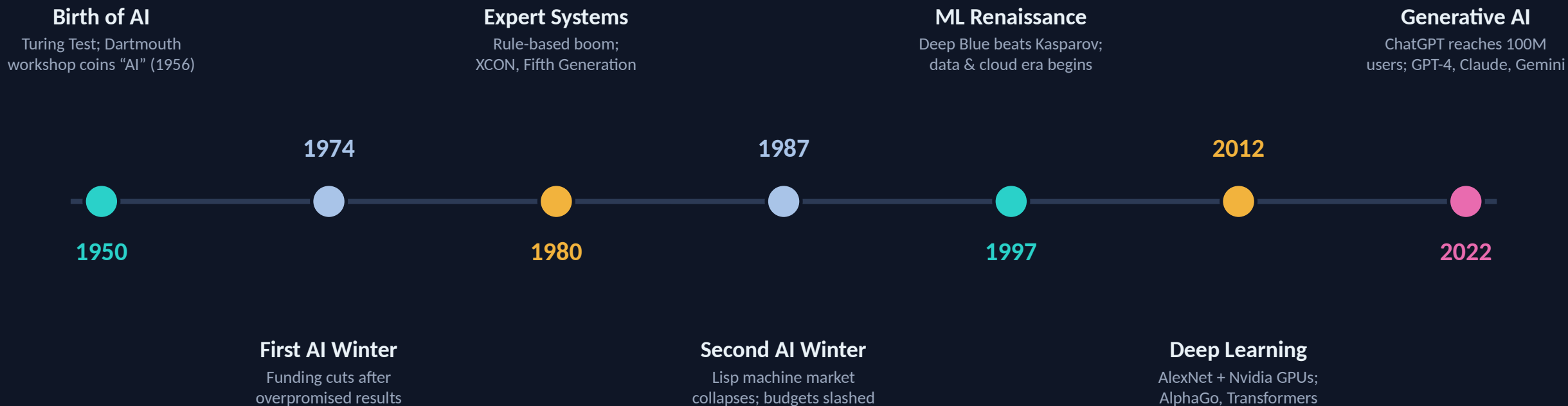
THE EVOLUTION OF ARTIFICIAL INTELLIGENCE

A Visual Timeline: 1950 – 2026

From the Turing Test to Generative AI — key milestones, AI winters, and the breakthroughs of deep learning and cloud computing.

Seven Decades at a Glance

The field advanced in waves — bursts of optimism, two long winters, then an explosion driven by data, GPUs, and the cloud.



The Birth of AI

1950 – 1969 • LOGIC, OPTIMISM, AND THE FIRST PROGRAMS

1950

Alan Turing asks “Can machines think?”

Turing publishes “Computing Machinery and Intelligence,” proposing the Imitation Game (Turing Test) as a practical way to judge machine intelligence.

1956

The Dartmouth Workshop — “Artificial Intelligence” is named

John McCarthy, Marvin Minsky, Claude Shannon, and others gather at Dartmouth College; McCarthy coins the term “artificial intelligence,” founding the field.

1957

Rosenblatt's Perceptron

Frank Rosenblatt builds the perceptron, an early trainable neural network — the conceptual ancestor of today's deep learning systems.

1966

ELIZA, the first chatbot

Joseph Weizenbaum's ELIZA imitates a psychotherapist using pattern matching, showing how easily people attribute understanding to machines.

1969

Shakey the Robot & the limits of perceptrons

SRI's Shakey combines perception, planning, and movement; the same year, Minsky and Papert's book Perceptrons exposes single-layer networks' limits, cooling enthusiasm.

Two AI Winters and the Expert-System Boom

1970 – 1995 • WHY PROGRESS STALLED — TWICE

1973

The Lighthill Report triggers the first freeze

A UK government review concludes AI failed to meet its grand promises; British and US (DARPA) funding is cut sharply.

1974–80

FIRST AI WINTER

Cause: overpromising + weak computers + perceptron criticism. Impact: labs close, research funding nearly disappears, and “AI” becomes a discredited label.

1980s

Expert systems revive the field

Rule-based programs like DEC's XCON save companies millions; Japan launches the Fifth Generation Computer project, and corporate AI spending booms.

1987–93

SECOND AI WINTER

Cause: the specialized Lisp-machine market collapses and expert systems prove brittle and costly to maintain. Impact: another decade of skepticism and slashed budgets.

Key lesson of the winters: hype that outruns hardware and data leads to collapse — a pattern researchers still reference today.

Machine Learning Rises — Data, Chips, and the Cloud

1997 – 2011 • THE QUIET COMEBACK

1997

IBM Deep Blue defeats Garry Kasparov

A machine beats the world chess champion for the first time — a symbolic proof that raw computation plus search can master a “human” domain.

2006

Deep learning is reborn — and the cloud arrives

Geoffrey Hinton shows how to train deep neural networks effectively; the same year, Amazon launches AWS, making massive computing power rentable to anyone.

2009

ImageNet: fuel for the revolution

Fei-Fei Li's team releases a dataset of millions of labeled images, giving neural networks the large-scale training data they had always lacked.

2011

IBM Watson wins Jeopardy!

Watson defeats champions Ken Jennings and Brad Rutter at natural-language trivia, bringing AI back into public view and corporate strategy.

The three missing ingredients — big data, cheap parallel compute, and better algorithms — finally converge in this period.

The Deep Learning Revolution

2012 – 2019 • WHEN HARDWARE CAUGHT UP WITH THEORY

2012

AlexNet — the breakthrough moment

Hinton's students Alex Krizhevsky and Ilya Sutskever crush the ImageNet contest using Nvidia GPUs, proving deep neural networks work at scale. The modern AI race begins.

2014

GANs invented

Ian Goodfellow introduces generative adversarial networks — machines that can create realistic images, opening the door to generative AI.

2016

AlphaGo defeats Lee Sedol

Google DeepMind's AlphaGo wins 4–1 at Go, a game once thought decades away from machine mastery, using deep reinforcement learning.

2017

“Attention Is All You Need” — the Transformer

Google researchers publish the Transformer architecture, the foundation of every modern large language model (the “T” in GPT).

2018

BERT and GPT-1 launch the LLM era

Google's BERT and OpenAI's first GPT show that pre-training giant models on internet text produces broad language ability.

The Generative AI Era

2020 – TODAY • AI GOES MAINSTREAM

2020

GPT-3 stuns researchers

OpenAI's 175-billion-parameter model writes essays, code, and poetry from plain prompts, revealing the power of scale.

2022

ChatGPT — the fastest-adopted app in history

Released in November 2022, ChatGPT reaches an estimated 100 million users in about two months, making AI a household topic overnight.

2023

GPT-4, Claude, Gemini — the frontier race

OpenAI (backed by Microsoft), Anthropic, and Google DeepMind compete on multimodal frontier models; Meta releases open-weight Llama models.

2024+

AI transforms industries — and Nvidia's chips power it all

AI spreads into medicine, finance, law, education, and software development; Nvidia's GPUs become the critical infrastructure, briefly making it the world's most valuable company.

Today's leading players: OpenAI, Google DeepMind, Anthropic, Meta AI, Microsoft, and Nvidia — with AGI as the openly stated goal for several of them.

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Research assistance: AI tools were used to help gather and organize milestone information; the timeline design and layout are my own original work.